

PRINCE ALBERT, SK, Canada

WMO: 718690

Lat: 53.2103N

Lon: 105.6767W

Elev: 428

StdP: 96.29

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.0	(c) -32.1	(d) -39.3	(e) 0.1	(f) -34.8	(g) -36.4	(h) 0.1	(i) -31.9	(j) 10.5	(k) -9.2	(l) 9.2	(m) -9.2	(n) 1.5	(o) 260	(p) 0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	28.8	19.1	27.0	18.2	25.5	17.1	20.5	26.6	19.3	25.2	18.2	23.6	4.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.8	23.5	17.1	12.8	21.9	16.1	12.1	20.9	61.0	27.0	56.8	25.1	53.4	23.6	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.3	(b) 8.3	(c) 7.5	(e) -39.1	(f) 32.4	(g) 3.3	(h) 1.9	(i) -41.5	(j) 33.8	(k) -43.4	(l) 34.9	(m) -45.3	(n) 36.0	(o) -47.6	(p) 37.4		
(a) 9.3	(b) 8.3	(c) 7.5	(e) -39.2	(f) 22.7	(g) 3.3	(h) 1.1	(i) -41.5	(j) 23.5	(k) -43.4	(l) 24.1	(m) -45.3	(n) 24.7	(o) -47.6	(p) 25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-16.7	-14.5	-7.3	2.7	10.1	15.3	18.1	16.8	11.3	3.4	-6.3	-14.2	(6)
(7)		DBStd	13.79	8.76	7.87	7.88	5.84	4.66	3.50	2.87	3.30	4.34	4.67	6.66	7.92	(7)
(8)		HDD10.0	3821	828	686	537	225	56	3	0	1	35	209	490	751	(8)
(9)		HDD18.3	6158	1086	919	795	469	256	101	39	70	212	462	740	1009	(9)
(10)		CDD10.0	771	0	0	0	6	61	163	251	210	76	6	0	0	(10)
(11)		CDD18.3	67	0	0	0	0	2	11	32	21	2	0	0	0	(11)
(12)		CDH23.3	910	0	0	0	3	58	161	341	280	64	2	0	0	(12)
(13)		CDH26.7	189	0	0	0	1	11	34	65	67	12	0	0	0	(13)
(14)	Wind	WSAvg	3.3	3.0	3.0	3.4	3.8	3.9	3.6	3.2	3.1	3.3	3.4	3.2	2.9	(14)
(15)	Precipitation	PrecAvg	426	17	14	18	26	43	74	75	56	38	26	17	21	(15)
(16)		PrecMax	599	36	45	57	66	150	163	179	197	107	92	43	40	(16)
(17)		PrecMin	279	3	1	2	0	5	7	20	13	2	1	2	8	(17)
(18)		PrecStd	73	8	9	11	18	32	37	38	40	26	20	9	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.4	5.0	10.6	22.1	28.0	30.2	30.8	31.2	28.6	21.4	10.4	4.9	(19)
(20)			MCWB	1.8	1.7	5.7	11.2	15.0	18.5	20.8	19.0	17.9	13.1	5.8	2.1	(20)
(21)		2%	DB	1.7	1.7	7.4	18.5	24.8	27.1	28.4	28.4	25.1	16.7	6.7	1.8	(21)
(22)			MCWB	-0.1	-0.4	3.4	9.5	13.7	17.2	19.9	18.9	15.5	10.1	3.7	0.0	(22)
(23)		5%	DB	-1.2	-0.7	5.2	15.6	22.4	25.0	26.8	26.4	22.4	13.8	4.2	-1.3	(23)
(24)			MCWB	-2.3	-2.1	2.1	8.2	12.6	15.9	18.8	18.0	14.6	8.4	2.0	-2.4	(24)
(25)		10%	DB	-4.0	-3.0	3.1	12.4	20.1	23.1	25.2	24.5	19.7	11.0	2.1	-3.3	(25)
(26)			MCWB	-4.8	-4.1	0.9	6.5	11.5	15.1	18.0	17.2	13.1	7.3	0.5	-4.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	2.0	6.0	12.0	16.6	20.4	22.7	21.4	18.7	13.7	6.4	2.4	(27)
(28)			MCDWB	4.1	4.6	10.1	20.0	23.8	27.5	28.2	27.9	26.0	20.7	10.0	4.7	(28)
(29)		2%	WB	0.1	-0.2	3.7	10.2	14.8	18.5	21.1	19.9	16.7	11.0	4.0	-0.1	(29)
(30)			MCDWB	1.6	1.5	6.9	17.0	22.8	24.5	27.0	26.4	23.0	14.9	6.4	1.6	(30)
(31)		5%	WB	-2.3	-2.0	2.3	8.6	13.4	17.4	19.8	18.7	15.0	9.0	2.2	-2.2	(31)
(32)			MCDWB	-1.2	-0.7	4.9	14.9	21.1	22.8	25.3	24.9	21.4	13.4	3.9	-1.2	(32)
(33)		10%	WB	-4.8	-4.1	0.9	6.9	12.0	16.2	18.7	17.7	13.6	7.3	0.5	-4.2	(33)
(34)			MCDWB	-4.1	-2.9	2.9	11.9	18.9	21.2	23.9	23.0	18.9	10.5	2.0	-3.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	11.3	11.0	11.7	13.5	11.8	11.6	12.3	12.3	10.0	8.2	9.0	(35)
(36)			MCDBR	12.4	12.4	12.2	17.0	17.6	15.6	14.5	15.9	17.6	15.6	10.9	11.2	(36)
(37)			MCWBR	10.9	10.5	8.8	9.6	8.7	7.4	7.4	7.8	9.3	9.7	8.2	9.9	(37)
(38)		5% WB	MCDBR	12.1	12.0	11.7	15.8	15.7	12.4	13.1	14.0	16.2	14.9	10.4	10.7	(38)
(39)			MCWBR	10.6	10.2	8.5	9.5	8.1	6.8	7.3	7.3	8.8	9.6	8.0	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.234	0.250	0.301	0.336	0.354	0.375	0.376	0.369	0.317	0.288	0.248	0.226	(40)
(41)		taud		2.319	2.316	2.265	2.360	2.367	2.351	2.336	2.363	2.508	2.542	2.453	2.331	(41)
(42)		Ebn at Noon		780	877	886	897	895	876	866	852	868	828	766	723	(42)
(43)		Edh at Noon		56	78	103	108	113	117	116	107	82	64	50	46	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.99	2.06	3.63	4.84	5.74	5.76	5.94	4.93	3.46	1.98	0.92	0.69	(44)
(45)		RadStd		0.10	0.12	0.26	0.42	0.50	0.44	0.38	0.40	0.35	0.32	0.13	0.09	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.38	+0.61	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page